

1. (7.5 points)

ABC Insurance is a commercial insurance company with complex agent/broker distribution channels. ABC Insurance is exploring the acquisition of Target X to increase its distribution flexibility.

Target X is an innovative mobile app based insurer that allows small business owners to customize and purchase general liability and property policies from their cellphones without the aid of an agent/broker.

Target X has consistently achieved a 15% ROE and 6% growth rate. Assume the risk free rate is 4%, the equity market risk premium is 5%, and company β is 1.10.

Target X has provided ABC Insurance with its abbreviated balance sheet as of December 31, 2017:

Assets (\$000)		Liabilities (\$000)	
Cash	17,250	Unpaid Claims	5,000
Investments	3,250	Unpaid Claims Expenses	1,250
Intangible Assets	2,000	Unearned Premium Reserves	8,500
All Other Assets	1,000	All Other Liabilities	2,750

ABC Insurance has obtained Price-Book Value multiples for similar, recent transactions across the industry:

Acquisition	ROE	Growth Rate	Discount Rate	Price-Book Value
#1	15%	6%	9.5%	1.00
#2	10%	3%	9.5%	1.25
#3	15%	6%	13.9%	1.50
#4	15%	6%	9.5%	1.70
#5	15%	6%	2.3%	1.75
#6	5%	-2%	2.3%	1.90

Target X also provided ABC Insurance with the following data from its most recent unpaid claims analysis, which utilized an over-dispersed Poisson (ODP) bootstrap model with five parameters:

ODP Bootstrap Model Fitted Incremental Paid Claims (\$000) as of (months)			
Accident Year	12	24	36
2015	2,703	1,297	1,000
2016	3,547	1,703	
2017	4,250		

Unscaled Pearson Residuals as of (months)			
Accident Year	12	24	36
2015	0.91	-1.31	0
2016	-0.79	1.15	
2017	0		

Standardized Pearson Residuals as of (months)			
Accident Year	12	24	36
2015	2.12	-2.12	0
2016	-2.12	2.12	
2017	0		

Random Sample of Standardized Pearson Residuals as of (months)			
Accident Year	12	24	36
2015	2.12	2.12	-2.12
2016	-2.12	-2.12	
2017	-2.12		

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1. (continued)

a. (1 point)

A type of strategic risk posed by the acquisition above is project risk - the risk of entering an acquisition without contemplating integration costs, timelines, reserve deficiencies, etc. Describe two additional types of strategic risk specific to the acquisition above.

b. (1 point)

A primary cause of property & casualty insurer insolvency is deficient loss reserves. Describe two additional potential sources of insolvency specific to the acquisition above.

c. (1.25 points)

Create a reasonable range of firm values for Target X using the Price-Book Value transaction multiples.

d. (2 points)

Calculate the mean of the accident year 2017 incremental paid between 24 and 36 months for the given random sample of standardized Pearson residuals.

e. (1 point)

Calculate the variance of the accident year 2017 incremental paid between 24 and 36 months for the given random sample of standardized Pearson residuals.

f. (0.5 point)

Describe how to incorporate process variance in the accident year 2017 incremental paid between 24 and 36 months for the given random sample of standardized Pearson residuals.

g. (0.5 point)

Assume Target X's claims settlement rate will change after the acquisition. Briefly describe how this may impact the process risk and parameter risk of Target X's unpaid claims model.

h. (0.25 point)

Assume Target X's actuarial department will operate independently after the acquisition. Briefly describe the internal systemic risk correlation between Target X and ABC Insurance.

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EXAM 7 SPRING 2018 SAMPLE ANSWERS AND EXAMINER'S REPORT

SPRING 2018 EXAM 7, QUESTION 1	
TOTAL POINT VALUE: 7.5	LEARNING OBJECTIVE(S): A5, A7, B3, C7
SAMPLE ANSWERS	
Part a: 1 point	
<u>Sample responses:</u> • Technology [or IT]: if another company patents the technology of buying insurance from a cell phone, then others will not be allowed to utilize this technology. • Reputation [or Brand] risk: collapse or erosion of reputation. Since it is a new distribution, many unforeseen problems would arise and damage ABC reputation. • Competition [or Competitor] risk: other competitors may enter the market using similar apps to offer lower prices. • Customers [or Client] and intermediary: the customers of ABC are used to having intermediaries such as agents or brokers to service them. With the acquisition, the core focus might be divided and agents/brokers unhappy as the new distribution bypasses them. • Industry risk: if the market is currently a soft market in the underwriting cycle, increasing market share will worsen the company performance. Industry risk is the risk within the industry that affects all companies within the industry. • Stagnation: currently, the firm has a 6% growth rate, but this might decrease to 0 in the next years, resulting to a stagnation.	
Part b: 1 point	
<u>Sample responses:</u> • Pricing inadequate → new line, so not much data → data limited → error in pricing → inadequate price → insolvency → Q.E.D. • Catastrophe Event: A significant catastrophe can easily generate large claims to an insurer and severely reduce its solvency. • Rapid Growth [or too fast growth]. Growing too fast can lead to deficiency, as won't have time to learn about your book. So this new book may have different development, different trends, so if grow too fast want to take those into account & will be mispriced for the losses coming in. • Reinsurance failure: not being able to recover ceded losses from the reinsurer may lead to insolvency in times of large catastrophes. • Underwriting: if ABC doesn't properly price target X's product (which essentially is new LOB for them i.e. inexperienced) may not have good uw results. • External Fraud: Since Target X is an app based insurer, ABC may lack expertise to detect claims fraud for claims filed on apps. • Overstated Assets: Target X will have provided overstated assets to ABC, which would skew ABC's perception of X thinking it is a good acquisition, even though in reality it is risky. • Reckless Management: Target X could have reckless management that may be making decisions that are not appropriate for firm's interests. • Insolvency of subsidiary: Groups can be rendered insolvent if one of their daughter companies goes insolvent.	

EXAM 7 SPRING 2018 SAMPLE ANSWERS AND EXAMINER'S REPORT

Part c: 1.25 points

Sample 1

Discount Rate $K = 0.04 + 1.1 \times 0.05 = 0.095$

Comparable transactions are #1 & #4

Equity for target X = Assets – Liabilities

$17,250 + 3,250 + 1,000 - 5,000 - 2,750 - 8,500 - 1,250 = 4,000$

Range of firm values = $[4,000 \times 1, 4,000 \times 1.7] = [4,000, 6,800]$

Sample 2

Discount Rate $K = 4\% + 1.1 \times 5\% = 9.5\%$

Comparable transactions are #1 & #4

Equity for target X = Assets – Liabilities, (excluding Intangible Assets)

$17,250 + 3,250 + 1,000 - (5,000 + 2,750 + 8,500 + 1,250) = 4,000$

Range of firm values = $[4,000 \times 1, 4,000 \times 1.7] = [4,000, 6,800]$

Part d: 2 points

Sample 1

Accident	12	24	36
<u>Year</u>	<u>Months</u>	<u>Months</u>	<u>Months</u>
2015	$2,813 = 2.12 \times \sqrt{2,703} + 2,703$	1,373	933
2016	3,421	1,616	
2017	4,112		

$$12 - 24 \text{ LDF} = \frac{2,813 + 1,373 + 3,421 + 1,616}{2,813 + 3,421} = 1.479$$

$$24 - 36 \text{ LDF} = \frac{2,813 + 1,373 + 933}{2,813 + 1,373} = 1.223$$

$$\text{Mean} = 4,112 \times 1.479 \times (1.223 - 1) = 1,356$$

Sample 2

Accident	12	24	36
<u>Year</u>	<u>Months</u>	<u>Months</u>	<u>Months</u>
2015	$2,813 = 2.12 \times \sqrt{2,703} + 2,703$	1,373	933
2016	3,421	1,616	
2017	4,112		

	12-24	24-36
LDF	1.48	1.22
% reported	0.553	0.818

EXAM 7 SPRING 2018 SAMPLE ANSWERS AND EXAMINER'S REPORT

$$Mean = \frac{4,112}{0.553} \times (1 - 0.818) = 1,356$$

Part e: 1 point

Sample 1

$$Var = \phi \times Mean$$

$$\frac{0.91^2 + (-1.31)^2 + 0^2 + (-0.79)^2 + 1.15^2 + 0^2}{6 - 5} = 4.491$$

$$Var = 4.991 \times \$1,356 = \$6,088$$

Sample 2

$$Var = \phi \times Mean$$

$$\phi = \frac{2.12^2 + (-2.12)^2 + 0^2 + (-2.12)^2 + 2.12^2 + 0^2}{6} = 2.996$$

$$Var = 2.996 \times \$1,356 = \$4,062$$

Part f: 0.5 point

Sample 1

We have calculated the mean in part (d) and variance in part (e). We can then match the moments to a gamma distribution and use a random number from the Gamma distribution to simulate an incremental paid loss value. This includes process variance.

Sample 2

Sample residuals from distribution of Gamma (m^{iter} , ϕm^{iter}) $\rightarrow$ Gamma(1356, 6088).

Sample 3

Simulate from gamma distribution $q_{2017,24-36} \rightarrow$ Gamma(mean=1356, var=6088).

Part g: 0.5 point

Sample responses for Parameter Risk

- Parameter Risk will increase as parameter estimate based on historical data does not reflect the current claims settlement pattern.

Sample responses for Process Risk

- Process risk is the pure effect of randomness associated with the insurance process. Since the change in settlement rate is not a change in the insurance process it does not impact this risk.
- Process risk will decrease if claim settlement rate increases as reserve will decrease, thus process variance will decrease.

EXAM 7 SPRING 2018 SAMPLE ANSWERS AND EXAMINER'S REPORT

Part h: 0.25 point
<u>Sample responses:</u> • Correlation will be low since they are eliminating the same actuary effect with different departments. • Data errors – if both companies rely on the same source data there could still be correlation due to data errors.
EXAMINER'S REPORT
Candidates were expected to tie in knowledge from multiple readings across all three Learning Objectives, including showing a basic understanding of strategic risks, insolvencies, standardized Pearson residuals, process variance, and internal systemic risks.
Part a
Candidates were expected to know types of strategic risks and be able to provide details related to the specific insurance situation of a merger and acquisition. Common mistakes included: • Citing plausible, real world strategic risks that were not able to be tied back to items provided in the stem of the question (e.g. catastrophe risk, estimation risk, parameter risk). • Identifying a subclass of the Project Risk given in the question. Discussions about IT integration, M&A problems, and cultural integration are all a subset within the process risk. Candidates were expected to identify uniquely different risks from process risks. • Naming a specific strategy risk while providing the definition for a different risk (e.g. listing Technology risk but then describing a competitor risk).
Part b
Candidates were expected to correctly identify and describe some of the primary reasons for impairments/insolvencies as mentioned in Brehm and provide details on the relevancy to the specific case. Common mistakes included: • Listing only one or no reason. • Listing strategic risks instead of primary risks of impairment/insolvency. • Providing reasons that did not describe how it related to impairment (e.g. rapid growth: rapid growth is a leading cause of impairment). • Listing risks that could happen, but were not identified as primary risks or substantially similar (e.g. Unearned Premium reserve problems, asset risk (but not overstated assets), latent claims, investment risk, M&A risk, parameter risk, estimation risk). • Duplicating risks, or using the same risk with different phrasing (e.g. Cat risk and Event risk; internal fraud and external fraud).
Part c
Candidates were expected to use the information provided to determine an appropriate range of values for a target acquisition by evaluating Target X against other recent transactions with the

EXAM 7 SPRING 2018 SAMPLE ANSWERS AND EXAMINER'S REPORT

same ROE, growth rate, and discount rate. Candidates were expected to calculate the book value of Target X and use the price to book value ratios to determine a reasonable range for the firm value using those pieces of information.

Common mistakes included:

- Not calculating the discount rate to use to evaluate acquisitions.
- Including intangible assets in the book value calculation.
- Not providing a range of values.

Part d

Candidates were expected to recognize the need to calculate the random sample incremental triangle using the random sample of residuals provided, and then calculate LDFs.

Common mistakes included:

- Not calculating the random sample triangle, instead using the given fitted claims to calculate the LDFs.
- Calculating only LDFs or incremental paid amounts using the given fitted triangle.
- Incorrectly calculating the random sample triangle using the standardized or unscaled residuals.
- Calculating the incremental paid for the wrong accident year or development period.

Part e

Candidates were expected to know the relationship $\text{Var} = \phi \times \text{Mean}$ for the ODP distribution, and the formula to calculate the scale factor, ϕ . There are two acceptable formulas provided for calculating ϕ in the syllabus; either was accepted:

$$\phi = \frac{\sum(\text{unscaled Pearson residuals})^2}{N - p}$$

$$\phi = \frac{\sum(\text{standardized Pearson residuals})^2}{N}$$

Common mistakes included:

- Mixing up the two formulas for ϕ .
- Trying to recalculate the unscaled Pearson residuals to solve for ϕ , rather than using the information provided. This approach doesn't work because the information needed to calculate the residuals is not provided in the problem.
- Using the wrong values of N (data points) or p (parameters) in the formula.

Part f

Candidates were expected to give a specific description of how to incorporate process variance by simulating incremental paid amounts from a distribution fit to the mean and variance calculated in parts (d) and (e).

EXAM 7 SPRING 2018 SAMPLE ANSWERS AND EXAMINER'S REPORT

Common mistakes included:

- Providing an answer that was too general, rather than for the specific accident year 2017 24-36 month increment asked in the question.
- Insufficiently explaining how the distribution would be utilized (e.g. simply saying “use a gamma(1356, 6084)”).

Part g

Candidates were expected to recognize that a change in settlement rate would impact parameter risk but not necessarily process risk.

Common mistakes included:

- Mixing up process and parameter risk.
- Stating that process risk will increase or decrease without providing support.

Part h

Candidates were expected to recognize that by operating independently the two departments would likely have a reduced correlation of internal systemic risk.

Common mistakes included:

- Listing external or macro risks as reasons for increased internal systemic risk.
- Listing various types of internal systemic risks without addressing the situation in question.